



IB CHEMISTRY INTERNAL ASSESSMENT FORM



WRCA

Student Name: **Hu, Jiarui (Jerry)**

Candidate Number:

0 5 1 5 5 8 0 0 0 0

TOTAL SCORE

7

Date Completed: **February 9, 2026**

RESEARCH DESIGN

Establish the scientific context for the work, state a clear and focused research question and use concepts and techniques appropriate to the Diploma Program level. Where appropriate, demonstrate an awareness of safety, environmental, and ethical considerations.

				SCORE
0	1 - 2	3 - 4	5 - 6	4
The student's report does not reach a standard described by the descriptors (right)	The research question is stated without context. <i>Methodological considerations</i> associated with collecting data relevant to the research question are stated . The <i>description of the methodology</i> for collecting or selecting data lacks the detail to allow for the investigation to be reproduced.	The research question is outlined within a broad context. <i>Methodological considerations</i> associated with collecting relevant and sufficient data to answer the research question are described . The <i>description of the methodology</i> for collecting or selecting data allows for the investigation to be reproduced with few ambiguities or omissions.	The research question is described within a specific and appropriate context. <i>Methodological considerations</i> associated with collecting relevant and sufficient data to answer the research question are explained . The <i>description of the methodology</i> for collecting or selecting data allows for the investigation to be reproduced.	

Clarifications for Research Design

Research Question:

- A research question with context should contain reference to the dependent and independent variables or two correlated variables, include a concise description of the system in which the research question is embedded, and include background theory of direct relevance.

Methodological considerations:

- the selection of the methods for measuring the dependent and independent variables
- the decisions regarding the scope, quantity and quality of measurements (e.g. the range, interval or frequency of the independent variable, repetition and precision of measurements)
- the identification of control variables and the choice of method of their control
- the recognition of any safety, ethical or environmental issues that needed to be taken into account.

Description of the Methodology:

- The description of the methodology refers to presenting sufficiently detailed information (such as specific materials used and precise procedural steps) while avoiding unnecessary or repetitive information, so that the reader may readily understand how the methodology was implemented and could in principle repeat the investigation.

MARKER'S COMMENTS

- The research question is placed into a well described context (with some error and omission)
(See red comment in ManageBac to improve this score)
- Methodological considerations are mentioned, but not discussed.
(See red comment in ManageBac to improve this score)
- The description of the methodology is excellent, with only a couple of ambiguities or omissions.

DATA ANALYSIS

Select, record, process and **interpret** data in a manner that is relevant to research question and supports the conclusion.

0	1 - 2	3 - 4	5 - 6	SCORE
The student's report does not reach a standard described by the descriptors (right)	The recording and processing of the data is communicated but is neither clear nor precise. The recording and processing of data shows limited evidence of the <i>consideration of uncertainties</i>. Some <i>processing of data</i> relevant to addressing the research question is carried out but with major omissions, inaccuracies or inconsistencies.	The <i>communication</i> of the recording and processing of the data is either clear or precise. The recording and processing of data shows evidence of a <i>consideration of uncertainties</i> but with some significant omissions or inaccuracies. The <i>processing of data</i> relevant to addressing the research question is carried out but with some significant omissions, inaccuracies or inconsistencies.	The <i>communication</i> of the recording and processing of the data is both clear and precise. The recording and processing of data shows evidence of an appropriate <i>consideration of uncertainties</i>. The <i>processing of data</i> relevant to addressing the research question is carried out appropriately and accurately.	3

Clarifications for Data Analysis

- Data refers to quantitative data or a combination of both quantitative and qualitative data.

Communication:

- Clear communication means that the method of processing can be understood easily.
- Precise communication refers to following conventions correctly, such as those relating to the annotation of graphs and tables or the use of units, decimal places and significant figures.

Consideration of Uncertainties:

- Consideration of uncertainties means that you are aware that uncertainty in measurements will affect your data processing and your results and you have taken appropriate steps to quantify the effect that uncertainty had on your investigation.

Processing of Data:

- Major omissions, inaccuracies or inconsistencies impede the possibility of drawing a valid conclusion that addresses the research question.
- Significant omissions, inaccuracies or inconsistencies allow the possibility of drawing a conclusion that addresses the research question but with some limit to its validity or detail.

MARKER'S COMMENTS

- The data is presented clearly and the processing is explained and well annotated
- The propagation of uncertainty is handled well, although initial measurement uncertainty is likely grossly underestimated.
- There are fundamental flaws in the application of the ideal gas law while processing the data that result in processed data that is incorrect.
- The selection of intervals for determining ΔP values is unjustified and makes no sense
(Fixing these will significantly improve your score)

CONCLUSION

Investigation and results discussion are linked with the research question and the accepted scientific context.

				SCORE
0	1 - 2	3 - 4	5 - 6	
The student's report does not reach a standard described by the descriptors (right)	A conclusion is stated that is relevant to the research question but is not supported by the analysis presented. The conclusion makes superficial comparison to the <i>accepted scientific context</i> .	A conclusion is described that is relevant to the research question but is not <i>fully consistent</i> with the analysis presented. A conclusion is described that makes some relevant comparison to the <i>accepted scientific context</i> .	A conclusion is justified that is relevant to the research question and <i>fully consistent</i> with the analysis presented. A conclusion is justified through relevant comparison to the <i>accepted scientific context</i> .	

Clarifications for Conclusion

Fully Consistent:

- A conclusion that is fully consistent requires the interpretation of processed data including associated uncertainties.

Accepted Scientific Context:

- Scientific context refers to information that could come from published material (paper or online), published values, course notes, textbooks or other outside sources. The citation of published materials must be sufficiently detailed to allow these sources to be traceable.

MARKER'S COMMENTS

- The entire conclusion was beyond the 3000 word limit.
- The conclusion is worth 25% of the IA score, but is only 6 sentences in a 24-page report
- It is hard for me to provide commentary on the conclusion knowing that so much of the data processing (and hence the analysis) is based on erroneous data.

EVALUATION

Examine and evaluate the appropriateness and quality of the investigation methodology and suggest improvements.

				SCORE
0	1 - 2	3 - 4	5 - 6	
The student's report does not reach a standard described by the descriptors (right)	The report states generic <i>methodological weaknesses or limitations</i>. Realistic <i>improvements</i> to the investigation are stated.	The report describes specific <i>methodological weaknesses or limitations</i>. Realistic <i>improvements</i> to the investigation that are relevant to the identified weaknesses or limitations, are described.	The report explains the relative impact of specific <i>methodological weaknesses or limitations</i>. Realistic <i>improvements</i> to the investigation, that are relevant to the identified weaknesses or limitations, are explained.	

Clarifications for evaluation

Methodological Weaknesses and Limitations:

- Generic is general to many methodologies and not specifically relevant to the methodology of the investigation being evaluated.
- Methodological refers to the overall approach to the investigation of the research question as well as procedural steps.
- Weaknesses could relate to issues regarding the control of variables, the precision of measurement or the variation in the data.
- Limitations could refer to how the conclusion is limited in scope by the range of the data collected, the confines of the system or the applicability of assumptions made.

Improvements

- Improvements must be directly related to the weaknesses and limitations that have just been discussed
- It must be clearly shown how the improvement addresses a specific weakness or limitation and it must be explained how that improvement will mitigate (lessen) or eliminate that specific weakness or limitation

MARKER'S COMMENTS

- The entire evaluation was beyond the 3000 word limit
- Only one 'limitation' is mentioned (but the student talks about it twice), and the student seems to have a fundamental misunderstanding about whether or not it was actually a limitation in the experimental design.